

ARRAY-11 ENGINEERING PACKET: OFFSHORE INTER-ARRAY CABLE LAYOUT

1. SCOPE

A 64-turbine offshore wind farm must be connected to an offshore substation platform by inter-array cables. Turbine positions are given exactly in the table that follows (integer metres). The two candidate platform positions, P (primary) and A (alternative), are shown on the site plan (Figure 1) and must be read from its gridlines. The available cable types, with their capacities and prices per metre, are given only by the cable catalogue chart (Figure 2). All costs are whole numbers.

2. LAYOUT RULES

A layout gives every turbine exactly one outgoing cable, to another turbine or to the scenario's platform, such that following outgoing cables from any turbine always reaches the platform; the cables therefore form trees rooted at the platform.

The load of a cable is the number of turbines whose power flows through it: the turbine it leaves plus every turbine upstream of that turbine.

Every cable is a straight segment between the two points it joins. Its length is the Euclidean distance in metres, rounded to the nearest whole metre with exact halves rounded up.

A cable between two turbines may be at most 1300 metres long. A cable from a turbine to the platform has no length limit.

No two cables may cross: two cables cross if their segments share any point other than an endpoint they have in common; a cable that touches another cable's interior, or two collinear cables that overlap along a stretch, also count as crossing.

At most B cables may end at the platform, where B is the scenario's number of feeder bays.

Each cable is built from exactly one cable type. It must use a type whose capacity is at least its load, and it always uses the cheapest such type available in the scenario. A cable's cost is its rounded length multiplied by that type's price per metre, and a layout's cost is the sum of its cable costs.

3. SCENARIOS

S1: platform P, all cable types in the catalogue, B = 6.

S2: platform P, all cable types, B = 5.

S3: platform P, every type except the smallest-capacity type, B = 6.

S4: platform A, all cable types, B = 6.

S5: platform A, all cable types, B = 5.

In every scenario only that scenario's platform exists; the other position is empty sea.

4. TURBINE COORDINATES (metres, exact)

T01	x =	4720	y =	-80	T33	x =	-120	y =	3060
T02	x =	3040	y =	-20	T34	x =	1700	y =	3080
T03	x =	1680	y =	20	T35	x =	3100	y =	3180
T04	x =	80	y =	40	T36	x =	4860	y =	3320
T05	x =	780	y =	340	T37	x =	700	y =	3520
T06	x =	5640	y =	340	T38	x =	2560	y =	3700
T07	x =	2400	y =	480	T39	x =	3980	y =	3700
T08	x =	3840	y =	520	T40	x =	5700	y =	3720
T09	x =	0	y =	660	T41	x =	4740	y =	3960
T10	x =	4700	y =	700	T42	x =	-120	y =	4000
T11	x =	1700	y =	820	T43	x =	1600	y =	4080
T12	x =	3260	y =	940	T44	x =	3220	y =	4120
T13	x =	3920	y =	1040	T45	x =	3880	y =	4360
T14	x =	700	y =	1060	T46	x =	760	y =	4380
T15	x =	2360	y =	1080	T47	x =	5440	y =	4380
T16	x =	5520	y =	1140	T48	x =	2440	y =	4480
T17	x =	1440	y =	1540	T49	x =	1580	y =	4720
T18	x =	-140	y =	1600	T50	x =	4920	y =	4760
T19	x =	3220	y =	1680	T51	x =	3100	y =	4780
T20	x =	4680	y =	1700	T52	x =	-60	y =	4940
T21	x =	2500	y =	1860	T53	x =	780	y =	5080
T22	x =	3920	y =	1900	T54	x =	4080	y =	5100
T23	x =	840	y =	1980	T55	x =	2540	y =	5240
T24	x =	5440	y =	2140	T56	x =	5740	y =	5280
T25	x =	1520	y =	2280	T57	x =	1540	y =	5520
T26	x =	80	y =	2380	T58	x =	3360	y =	5540
T27	x =	3300	y =	2400	T59	x =	60	y =	5600
T28	x =	4920	y =	2440	T60	x =	4660	y =	5760
T29	x =	3840	y =	2640	T61	x =	860	y =	5840
T30	x =	640	y =	2680	T62	x =	4100	y =	5880
T31	x =	5540	y =	2720	T63	x =	5660	y =	5960
T32	x =	2520	y =	2880	T64	x =	2320	y =	6160

Site plan. P = primary platform, A = alternative platform (square centre, marked by the cross hair). Turbine dots are for orientation only: use the table.

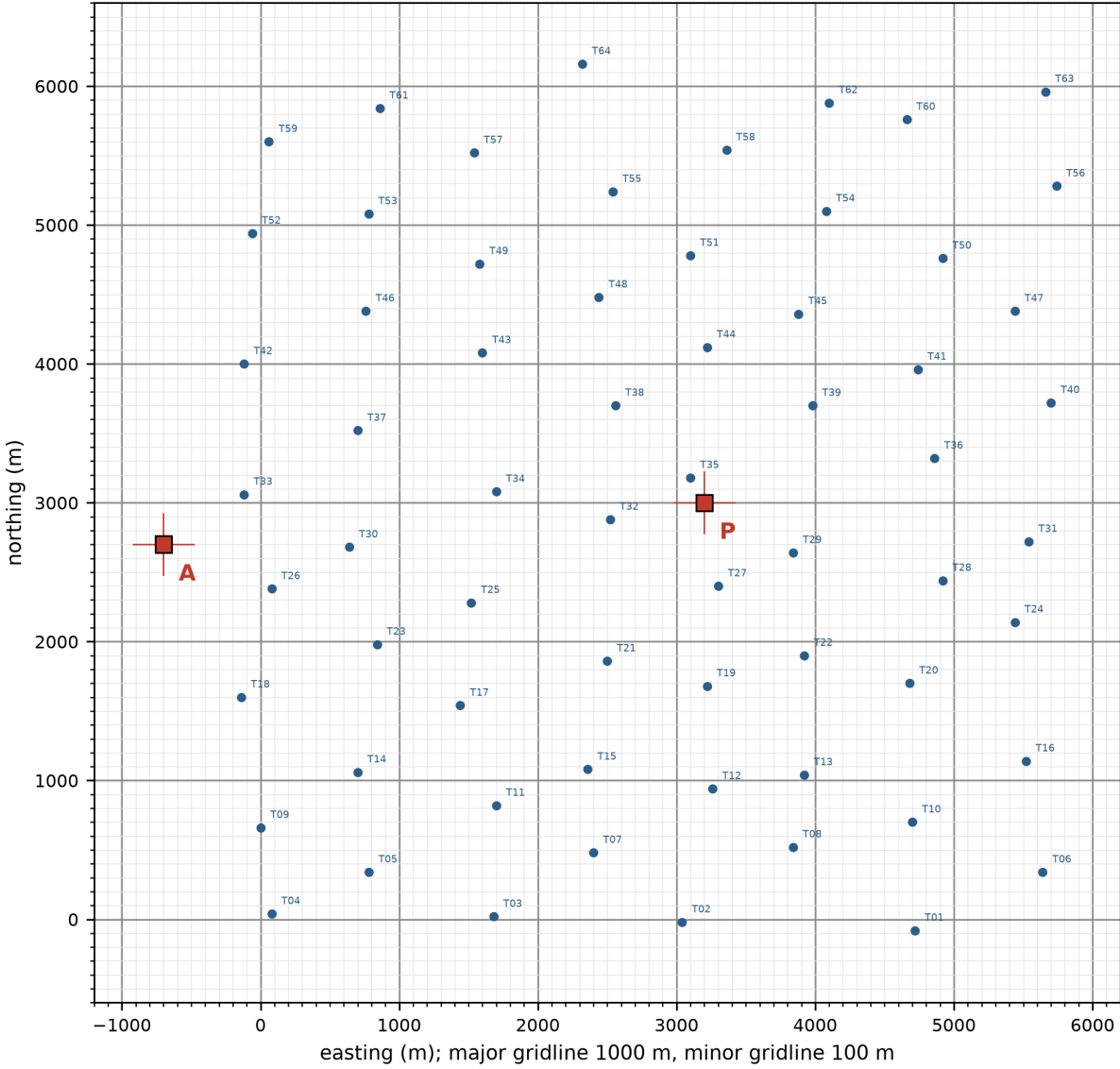


Figure 1 -- Site plan.

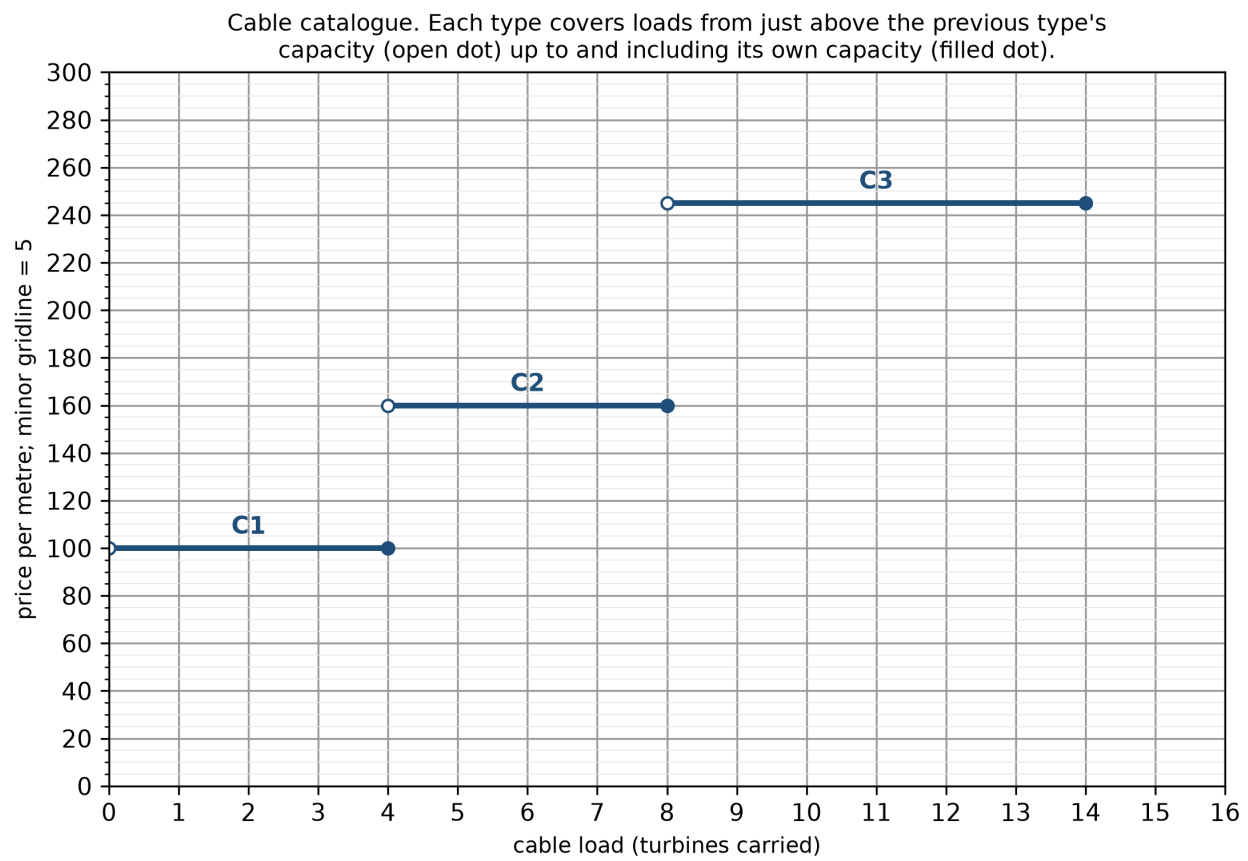


Figure 2 -- Cable catalogue.